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SECJ3553 ARTIFICIAL INTELLIGENCE

SECTION 02

Assignment 2

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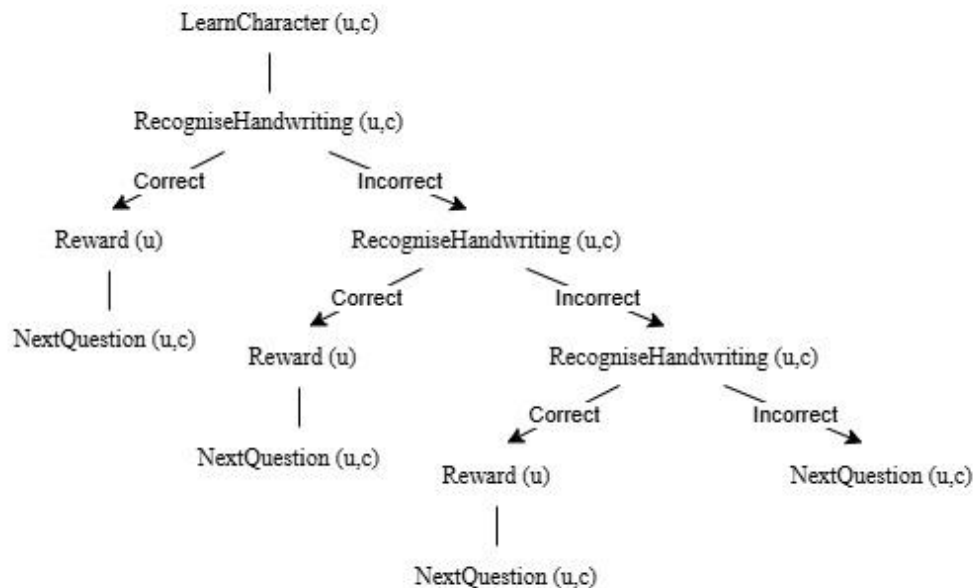
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Introduction

The state space diagram represents the states and actions involved when the user is learning a character, from the selection of the character to be learned, until the character's learning process is complete.

State Space Diagram



Description

The characters are labelled as follows for easier differentiation of character status.

Characters Labelling

- “Not Learned”: The user has not been introduced to the character
- “Learning”: The user is in the process of learning the character and has not yet mastered it
- “Mastered”: The user has mastered the character with at least 90% accuracy in the last 5 attempts

State	Action
LearnCharacter Determine whether to introduce a new character or revise an already learned character.	At the beginning of the process, a character will be selected to be learned in this cycle. The selected character may either be a new character that the user is learning or a newly introduced character. The character to be selected will depend on the number of

	unmastered characters and the accuracy of the learned characters. First, the system will determine whether a new character should be introduced. If the number of characters labelled as "Learning" is less than the threshold of 5, a new character labelled as "Not Learned" will be introduced. If the number of characters labelled as "Learning" is greater than or equal to the threshold of 5, a character labeled as "Learning" or "Mastered" will be presented. The selected character will be determined based on the mastery level of that character and the interval of revision.
RecogniseHandwriting This state evaluates the user's handwritten Mandarin characters to determine correctness.	In this state, each attempt is assessed based on stroke order, stroke direction and overall similarity to the target character. Every attempt, whether correct or incorrect, updates the character's success and error rates. Repeated incorrect attempts increase the error rate. Once the error rate exceeds a defined threshold, the character will appear more frequently in future exercises to reinforce mastery. The user also receives AI-generated feedback highlighting whether and how their handwriting and performance can be improved.
Reward This state rewards the user if good learning behavior is detected.	There are several situations in which users can be rewarded, including maintaining a continuous learning streak over consecutive days and showing a progress in learning. In addition, users will also be rewarded for solving difficult problems (where the overall error rate of the word is high) or achieving high accuracy in revision questions. When a user is close to reaching a reward state, the system will provide positive feedback to encourage continued engagement. This state motivates users to keep learning until they reach a certain proficiency level and become more skilled in the language.

Goal: NextQuestion

- Which denotes the conclusion of the learning cycle for a chosen character c by user u is the objective state of this state space problem. Once the system has fully assessed the user's engagement with the character through handwriting recognition and where applicable, rewards are provided.
- When the system reaches the NextQuestion state, it has acquired enough data regarding the user's performance, including accuracy, error rate and learning progress. Based on the user's learning history and current mastery level, the system then moves on to presenting the next suitable character or revision work.

- In order to retain user involvement and reinforce effective learning behavior, this goal state guarantees that the learning process is continuous and flexible, enabling the system to dynamically choose the next character.

Path Cost

The minimum path cost shows the least number of checks needed to validate a character. In this case, the user learns the character and gets it right on the first handwriting attempt. Because the answer is correct, the system gives a reward and then moves on to the next question.

The maximum path cost shows the greatest number of checks needed before the system moves on. Here, the user learns the character but fails three consecutive handwriting recognition attempts. Since all checks are incorrect, the system skips the reward and moves directly to the next question.

Solution

A valid solution path is any path from the initial state to the goal state, $\text{NextQuestion}(u, c)$.

- **Correct on first attempt**

$\text{LearnCharacter}(u, c) \rightarrow \text{RecogniseHandwriting}(u, c) = \text{correct} \rightarrow \text{Reward}(u) \rightarrow \text{NextQuestion}(u, c)$

- **Correct on second attempt**

$\text{LearnCharacter}(u, c) \rightarrow \text{RecogniseHandwriting}(u, c) = \text{incorrect} \rightarrow \text{RecogniseHandwriting}(u, c) = \text{correct} \rightarrow \text{Reward}(u) \rightarrow \text{NextQuestion}(u, c)$

- **Correct on third attempt**

$\text{LearnCharacter}(u, c) \rightarrow \text{RecogniseHandwriting}(u, c) = \text{incorrect} \rightarrow \text{RecogniseHandwriting}(u, c) = \text{incorrect} \rightarrow \text{RecogniseHandwriting}(u, c) = \text{correct} \rightarrow \text{Reward}(u) \rightarrow \text{NextQuestion}(u, c)$